

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022

Course: B. Tech.

Branch : Computer Science and Engineering

Semester : VII

Subject Code & Name: BTCOE704(A) Blockchain Technology

Max Marks: 60

Date: 07/02/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. Use of non-programmable scientific calculators is allowed.
3. Assume suitable data wherever necessary and mention it clearly.

	Marks
Q. 1 Solve Any Two of the following.	12
A) What is Blockchain? Explain key elements of it.	6
B) Difference between public vs. private Blockchain.	6
C) Explain Merkel Tree and Merkel Tree Structure.	6
Q.2 Solve Any Two of the following.	12
A) What is Bitcoin Mining and How Does Bitcoin Mining Work?	6
B) Why Proof of Burn required and how does algorithm work?	6
C) How do P2P networks work? Explain its type.	6
Q. 3 Solve Any Two of the following.	12
A) Explain design issues for permissioned Blockchain with any 2 issues.	6
B) Explain Paxos algorithm in detail.	6
C) What is Byzantine general problem and How Bitcoin Solves the Byzantine General's Problem?	6
Q.4 Solve Any Two of the following.	12
A) How KYC on Blockchain can help?	6
B) How Blockchain benefits food security in future?	6
C) How does Digital Identity work today using Blockchain?	6
Q. 5 Solve Any Two of the following.	12
A) Explain Hyperledger Fabric and how does it work?	6
B) What are the benefits of Hyperledger Fabric?	6
C) Explain writing smart contract using Ethereum ?	6

***** End *****